

Are we alone *for now*? A frequentist perspective on Fermi’s Paradox and the existence of extraterrestrial life

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Abstract

The statistical treatment of uncertainty in astrobiology has been largely dominated by Bayesian methods. This paper argues this dominance is grounded in a misunderstanding of the link between the philosophy of uncertainty (statistics) and observation of cosmological phenomena and is therefore morally unwarranted. We propose the “fixed observer” hypothesis; in particular, we claim human observational access to the universe is doubly constrained, both by the speed of light and the cosmic blip of human existence, and thus effectively fixed. Given these constraints, the factors characterizing the presence of extraterrestrial life are not random parameters endowed with prior distributions but instead fixed physical constants about which we have severe epistemic uncertainty. The appropriate inferential framework is then the frequentist one, assigning probabilities to procedures rather than to parameters and preserving the distinction between our not knowing of a fixed quantity and the genuine randomness of a stochastic process.

1 Introduction

The question of whether life, much less intelligent life, exists elsewhere in the universe is one of the most imaginative and intriguing questions in science, yet it is one for which we possess almost no direct evidence to answer. To this end, the statistical philosophy we use to describe and discuss uncertainty carries an enormous intellectual burden: both as a framework for computation and a set of assumptions about what our observations can reveal to us about what is not known. Across the recent literature in astrobiology and observational astronomy, that burden has been borne largely by Bayesian methods, being treated not as one reasonable choice among several but as the natural and appropriate language for reasoning under uncertainty (e.g., Verendel and Häggström 2017; Catling et al. 2018; Lingam, Nichols, and Balbi 2024). We argue the dominance of these Bayesian methods is morally weak and rests on a misunderstanding of the relationship between statistical uncertainty and the specific conditions under which cosmological observation takes place.

An important distinction is that this misunderstanding is not technical but philosophical. What the Bayesian framework requires (a prior probability distribution over parameters of interest) is only coherent if those parameters are, in some meaningful way, variable. We contend the parameters governing the existence or, if existent, prevalence of extraterrestrial life are non-random. They are instead fixed physical constants out of observational

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reach, and thus, our uncertainty about them is purely epistemic: a consequence of incredibly sparse data and not of genuine stochastic variation. This distinction, between knowing the value of a fixed quantity and observing a quantity that genuinely varies, is one that the existing literature overlooks. Whether that oversight reflects a deliberate pragmatism—a willingness to reach for a framework producing tractable and informative to a seemingly unanswerable question—or simply the unchallenged inheritance of methods proven powerful elsewhere in the hard sciences, the effect is the same: the distinction is assumed away rather than reasoned through. We argue this distinction cannot, in good faith, be assumed away without consequence.

In what follows, we provide an introduction to the two dominant camps of modern statistical thought: frequentist and Bayesian statistics. Section 2 introduces the “Fixed Observer” hypothesis. Section 3 makes the case for frequentist methods in astrobiology and other fields attempting to address Fermi’s Paradox, while Section 4 presents a more thorough critique of Bayesian methodology in astronomy and astrobiology. Section 5 concludes.

1.1 Philosophical Perspectives on Uncertainty

Philosophically, there are two different perspectives to characterizing uncertainty: the frequentist and Bayesian frameworks. In astronomy and astrobiology, fields which often demand a link between large-sample and $n = 1$ unique events, selecting the “appropriate” statistical framework plays a critical role in our capacity to answer questions like “where are they?” or “are we inevitable?” Indeed, in fields where our entire base of prior knowledge consists of one, evolutionary contingent inhabited planet, observed from a single location in space-time, this disagreement, and therefore, choice, determines what our data can and cannot tell us.

In frequentist (also called classical) statistics, a parameter θ , such as the fraction of stars with planetary systems f_p , is an immutable, fixed constant. We characterize our knowledge of the parameter (supposing it to be unknown) by our observation of data generated under it. We do not make probabilistic statements about θ itself, but rather about our estimators and testing procedures across (often hypothetical or simulated) repetitions of measurement. The branch of frequentist statistics most comparable (granted, as comparable as an apple is to a pear) to the Bayesian position is conditional inference, developed first by R.A. Fisher and formalized by Cox (1958). Cox posits that when data can be decomposed into two components, a sufficient statistic $T(x)$ capturing all available information about θ and an ancillary statistic $A(x)$ whose distribution does not depend on θ but carries information about the precision or configuration of the measurement, then inference should be conducted conditionally on the observed value of the ancillary statistic.

Cox’s (1958) mixture experiment illustrates this point simply: suppose a scientist randomly selects one of two instruments of significantly different levels of precision before taking a reading of some measure. The unconditional frequentist analysis, averaging over both instruments, yields a reference distribution corresponding to neither the data actu-

ally obtained nor any other that could arise from a single well-defined measurement. It is, in a more intuitive sense, an answer to a question that was never asked. Conditioning on which instrument was actually used (conditioning on the ancillary) recovers a reference distribution genuinely relevant to the question at hand. The parameter θ remains fixed throughout; what changes is the reference set over which the procedure's behavior is evaluated.

Conditional inference is the frequentist procedure most analogous to what Bayesian inference attempts. Both ask, in some sense, "*given what we actually observed*, what can we conclude?" But the analogy breaks down at the point that matters for astrobiology. The frequentist conditions on the ancillary (a feature of the data uninformative about θ but defines the relevant measurement context) and continues to treat θ itself as a fixed unknown. The conditioned sampling distribution remains a distribution over hypothetical realizations of the data. The structure is reversed under Bayesian inference: the parameter θ is treated as a random variable governed by a prior $\pi(\theta)$ encoding the researcher's prior knowledge or beliefs, and Bayes' Theorem yields the posterior $\pi(\theta|X) \propto L(X|\theta)\pi(\theta)$ for observed data X .

It is worth acknowledging the genuine appeal of this framework when a well-founded prior is available. An informative prior, one constructed from genuine domain knowledge, previous experiments or theoretical constraints, can meaningfully sharpen inference in data-sparse settings by encoding what is already known before a new observation is taken. In a field where making new observations is costly or when droves of experimental or observational records are available, it is, perhaps, no surprise Bayesian methods have found considerable popularity. In these settings, the posterior is a principled synthesis of prior knowledge and new data, and the Bayesian framework is earning its precision rather than manufacturing it.

Difficulties arise when no such prior is available: cornering the researcher to fall back on sets of uninformative or naïve priors chosen for their mathematical convenience, neutrality, or conservativeness. When the likelihood is too weak to dominate the prior (that is, when the data are too sparse to meaningfully change our existing knowledge), as it invariably happens to be when our prior knowledge comes from a single planet inhabited by a "special" form of life, the posterior remains the prior.

While far from a scientific law and at best a "wonderful way to organize our ignorance," (Rhodes and Tarter 1999), the Drake equation and its parameters demonstrate the weakness of the Bayesian approach cleanly. The parameters of the Drake equation are those for which we have no prior experiments, theoretical derivations from first principles, nor analogous systems from which to borrow to inform our estimates. Every published Bayesian treatment of these parameters is, whether its authors acknowledge it or not, fundamentally operating in the uninformative regime. The apparent informativeness of the resulting posteriors is inherited from the prior specification, not extracted from data.

1.2 Uncertainty in Science

The relevant structural issue is the distinction between aleatory and epistemic uncertainty. Der Kiureghian and Ditlevsen (2009) draw this distinction with precision. Aleatory uncertainty is inherent, irreducible randomness in a physical system (e.g., the quantum uncertainty in a radioactive decay time, the thermal noise in a detector unit, etc.). Epistemic uncertainty is uncertainty arising from incomplete knowledge, reducible in principle by gathering more data or deepening theoretical understanding. The distinction matters practically because conflating the two introduces artificial correlation in probabilistic models and produces miscalibration of uncertainty estimates. As Aldous (2018) observes, classical frequentist statistics implicitly maintains the distinction: the likelihood function captures aleatory variation in the data, while the parameters represent fixed unknowns, the objects of epistemic uncertainty. A Bayesian who places a prior distribution on the parameters implicitly collapses the distinction, encoding ignorance about a fixed quantity in the mathematical form of a random variable. For the physical quantities of interest in astrobiology (e.g., stellar formation rates, planetary habitability fractions, the intrinsic rate of abiogenesis) our uncertainty about them is purely epistemic. We have not measured nor observed them precisely enough not because they genuinely vary but because our observations are sparse.

The frequentist response is that domain knowledge ought to enter statistical analyses as structure rather than as probability distributions over parameters. The preference is to avoid formal prior probability models not because domain knowledge is irrelevant but because encoding it as a prior conflates (or more cynically, ignores) two qualitatively distinct epistemic acts: understanding the structure of the problem (a scientific judgement) and quantifying uncertainty about a specific parameter value (a statistical task).

Sober (2008) adds a dimension the formal statistics literature tends to overlook. Scientific inference is not "personal belief updating," but an enterprise in which claims must be both communicated and justified across investigators and institutions with different prior beliefs. A posterior distribution whose width and location are substantially determined by a prior (i.e., when the data are insufficient to update the prior) is not, in Sober's eyes, a scientific result but rather a quantitative expression of the researcher's private beliefs formatted in the language of probability.¹ Bayesian inference over competing hypotheses requires assigning a likelihood to the complement of the hypothesis under consideration—that is, every alternative not explicitly modeled. This "catch-all problem," that the complement cannot be assigned any objective probability because it includes hypotheses which have not yet been conceived, is a disqualifying blunder of Bayesian methods in astrobiology. The space of possible habitable planets, biochemistries, and evolutionary trajectories consistent with "life exists elsewhere" is almost entirely unconceived.

¹I suspect this to be an incredibly controversial take; I pray the Bayesians do not crucify me for it.

2 The “Fixed Observer” Hypothesis

The cosmological constraints on human observation place us at what is, for all practical purposes of statistical inference about the universe, a fixed point in time and space. We develop this “Fixed Observer” hypothesis in three steps: the hard boundary imposed by the cosmological light cone, the temporal fixity imposed by the relative brevity of human existence, and the structural constraint imposed by the anthropic selection effect. Together they establish the Drake equation factors are fixed, unknown constants at the edge of our observational horizon, not random variables, and that the appropriate framework is consequentially frequentist.

Consider first the light-cone constrain. The finite speed of light, $c \approx 3 \times 10^8$ meters per pecond, and the geometry of general relativity jointly impose a hard boundary on what any observer at our location in the universe can ever observe or measure. Davis and Lineweaver (2004) provide a more precise characterization: the particle horizon (the maximum distance from which we could have received any signal since the Big Bang) is currently 46.5 some billion light years in comoving coordinates, despite the universe being only 13.8 billion years old. The cosmic event horizon, which limits what we can ever observe in the future regardless of how long we wait, lies at approximately 16 million light years. Everything we can ever know about the universe, through any conceivable observation with any conceivable instrument, lies within this bounded region. Beyond it, the parameters governing the physics and biology of the universe are unfalsifiable and thus unknowable for any observer at our position and time. Bayesian inference placing a prior distribution over the number of civilizations in the full universe is assigning probability to claims no observation from our light cone could ever constrain. This is not statistics in any rigorous sense; it is metaphysics wearing the trench coat and hat of probability theory.

The temporal constraint is additionally restrictive. *Homo sapiens* have only existed for some 200,000 years— $\approx 0.001\%$ the age of the universe—and the genus *Homo* has been around for some 2.8 million years— $\approx 0.2\%$ the age of the universe. The earliest technologies permitting our deep study of space (e.g., Galileo’s telescope in 1609) have been around for what is the cosmological equivalent of less than a fraction of a millisecond ($\approx 2.9 \times 10^{-9}\%$ the age of the universe), and technologies enabling our direct exploration of the stars have been around for a share of time which is effectively zero ($\approx 4.3 \times 10^{10}\%$ the age of the universe).

The parameters we seek to estimate through the Drake equation describe the state of the universe’s large-scale processes at our cosmic epoch. Over the timescale of human existence, the universe’s large-scale properties (e.g., stellar formation rates, the distribution of planetary systems, chemical abundances governing biochemistry/ies) have changed by amounts negligible relative to our measurement uncertainties. While the universe is expanding, we are, in all practical respects, taking a single snapshot: a measurement at a relatively fixed moment in the universe’s developmental history. The Drake factors are the values of physical and biological constants at that moment. They do not vary across the hypothetical repetitions of our measurement but instead across cosmic time, on timescales

that dwarf our observational window by nine or ten orders of magnitude. This temporal fixity is the physical foundation of the random data-fixed parameter framework frequentists presuppose and Bayesians abandon by assigning prior distributions to what are, from our place in spacetime, deterministic.

These two observations—that the spatial boundedness of the light cone and the temporal fixity of our position in spacetime—jointly identify the structure of the inference problem. The Drake factors are fixed unknowns; our measurements are noisy estimates of data produced under those unknown fixed values; and our statistical task is to characterize uncertainty about those fixed values through procedures whose error properties are known and controlled. The Bayesian framework, by assigning priors to the Drake factors, treats what is a fixed parameter estimation problem as if it were a random variable inference problem. As we discuss later, the two regimes are not interchangeable, and the consequences of the substitution (prior dominance in a data-sparse regime) are the target of Section 3.

The anthropic constraint reinforces the argument from a distinct direction introducing additional reason why the Bayesian reasoning is poorly suited to the problems of astrobiology. Carter (1974)’s Weak Anthropic Principle—a generalizable modification to the Copernican principle that humans are not privileged observers of the universe—contends our location in the universe is necessarily privileged to the extent required by our existence as observers, and this extends to our position in time. We cannot be observing the universe from a point at which conditions for observers of our kind did not hold (put most simply, we are quite special). Our observation is therefore not a random draw from the full distribution of possible observational outcomes but a selected draw filtered by the hard condition (or luck) of our existence and survival. This selection, as is generally the case, has direct consequences for inference: the data generated by our special position (e.g., the fact life, much less an “intelligent” form of life capable of observing and asking these questions, arose on Earth) are not randomly sampled from the full space of planetary histories and thus only compatible with our unique experience as observers at our point in spacetime.

3 The Frequentist Case in Astrobiology

Our critique predicts Bayesian inference on the Drake equation and other topics in astrobiology should be prior-dominated due to data being too sparse to update any prior. Indeed, we find evidence supporting this prediction in the empirical literature. We examine four treatments in turn, ordered by the specificity of the epistemic error each commits: prior sensitivity in single parameter estimation (Spiegel and Turner 2012), parametric assumptions across the full Drake Equation (Sandberg, Drexler, and Ord 2018), categorical conversion of fixed constants into random variables (Maccone 2010), and a partial exception illustrating by contrast what a more feasible frequentist approach may look like (Frank and Sullivan 2016).

Spiegel and Turner (2012) provide a case study of parameter estimate sensitivity to prior specification. They model the abiogenesis rate as following a Poisson process with parameter λ and compute Bayesian posteriors given Earth’s relatively early emergence of life at 700 million years after planetary formation. Three uninformative priors yields three qualitatively different conclusions. A uniform- λ prior implies life probably common; a log-uniform prior is ambiguous; and an inverse prior (λ^{-1}) implies life is probably rare. After incorporating the anthropic selection correction—that we can only be asking this question on a planet where we know life arose—their conclusion is ambiguous: Earth’s early biogenesis is consistent with an arbitrarily low abiogenesis probability for any of the prior choices. As predicted, the data are insufficiently informative to distinguish between prior choices, and the posterior inherits the prior’s commitments.

The appropriate frequentist response to the same data is different in a revealing way. Supposing we accept the parametric Poisson model Spiegel and Turner posit, a frequentist instead reports a confidence interval from the model with one observed event. The conclusion, to no surprise, is the data provide insufficient power to exclude any abiogenesis rate above a lower bound. The resulting interval spans many orders of magnitude, and that width is not a failure of the analysis but an honest quantitative expression of the evidence that one observation from a Poisson process constrains the rate parameter only weakly from above and barely from below. Granted, the Bayesian posterior can be artificially narrowed by specifying a more concentrated or informative prior, creating the appearance of precision or insight the data do not support. The frequentist interval, conversely, cannot be narrowed without additional data.

A generalization to the work Spiegel and Turner comes from Sandberg et al. (2018). Assigning a probability distribution to each Drake factor and log-uniform priors spanning many orders of magnitude to the biological parameters f_ℓ, f_i, f_c , Sandberg and coauthors use Monte Carlo simulations to obtain a distribution over N , the number of communicating civilizations. The acknowledge naive point estimates of N are misleading because the product of many uncertain factors is log-normally distributed, and the mean of such a product can be enormous even when the median is near zero. Still, their analysis requires the pre-specification of probability distributions for individual factors of the Drake equation weakly derived, if at all, from data. A log-uniform distribution spanning twelve orders of magnitude for the abiogenesis fraction is less a scientific finding than an admission of complete ignorance. Indeed, the wide interval for Sandberg et al.’s probability of cosmic solitude (38-85%) is, as in the case for Spiegel and Turner’s estimates, inherited by the choice of distributional families for the biological parameters and weakly dependent on observed data.

The clearest illustration of the categorical error underlying both approaches is presented by Maccone (2010). He treats each of the seven factors in the Drake equation as a positive random variable drawn independent from a distribution chosen by the researcher, invokes the Central Limit Theorem to establish the logarithm of the product is approximately distributed normally, and concludes the number of communicating civilizations

N follows a lognormal distribution, reporting a mean N of approximately 4,590 with a spread of 3,953. But Maccone’s calculations are only as correct as his assumptions, which are overly parametric and misinterpret the frequentist approach to statistics. As we have argued, the Drake factors are not random variables but fixed constants about which we have epistemic uncertainty. Maccone’s log-normal is a model of his uncertainty distribution over the space of possible universes compatible with his prior beliefs rather than a model of our particular universe. This distinction matters because the distribution he computes can generate a mean simply by adjusting the width of the input priors: wider priors produce higher means, not because the universe is richer, but because more probability mass is placed on higher values of N . A conspicuous frequentist treatment would report intervals and estimates dependent on the data, which we know to be far less numerically precise given the sparsity of currently available data, and therefore more epistemically honest.

Frank and Sullivan (2016) offer a partial and instructive exception. They reformulate the Drake equation to eliminate its most uncertain parameters—the longevity of communicating civilizations L —by asking instead how many technological species have ever arisen in cosmic history. This approach instead yields a threshold condition: unless the combined biotechnical probability f_t^b is below approximately 10^{-24} , humanity is not unique in the history of the observable universe. In identifying a specific decision threshold and asking whether current astronomical data rule out the region below it, Frank and Sullivan’s framing is frequentist in flavor. Indeed, because the threshold lies so far below any plausible estimate, the conclusion follows with negligible sensitivity to distributional assumptions. Frank and Sullivan’s approach is more transparent and more honest than its Bayesian counterparts because it asks only what the data can answer rather than manufacturing an inferential machinery elaborate enough to produce an answer to questions the data cannot reach.

4 The Structural and Empirical Failure of Bayesian Methodology

The foundational statement of the Bayesian case for astrophysics is Loredo (1992), who catalogued common errors in frequentist astronomical practice (e.g., misinterpretation of confidence intervals, incorrect treatment of upper limits, improper marginalization over nuisance parameters) and argues Bayesian methods resolve each. His position is, for the specific technical problems he identifies in well-specified physical models, largely correct. But Loredo’s framing contains a critical admission too easily passed over in the three decades of Bayesian dominance in astrophysics and astrobiology. He writes that in a strict frequentist view, for example, it is meaningless to speak of the probability of the true flux of a star: the true flux is, by definition, a single fixed value. He offers this as a limitation of frequentism. We argue the opposite: it is an accurate and precise description of the distinction we repeatedly draw throughout the paper: the true abiogenesis rate on a habitable planet is a single fixed value in exactly the same sense as the true flux of a star.

When Loredó’s own defense of Bayesian astrophysics acknowledges the fixed-parameter framework and proceeds to assign prior distributions to those fixed parameters anyway, the move requires justification the text does not supply. If the frequentist is right the true flux is a fixed value, the Bayesian prior over that flux is a shoddy objectification of the researcher’s beliefs about a fixed value. By comparison, in data-rich settings, the data overwhelm those beliefs and the difference is inconsequential. In astrobiology, they do not.

Empirical evidence for the consequences of this inconsistency comes from Tak, Ghosh, and Ellis (2018), who reviewed 75 published papers in the *Astrophysical Journal* and *Monthly Notices of the Royal Astronomical Society* and found more than 30% of them used Bayesian analyses without rigorously verifying “posterior propriety” (that is, without confirming the posterior distribution integrates to a finite constant and therefore constitutes a valid probability measure). This matters because Monte Carlo algorithms can produce numerical output that appears well-behaved even when sampling from an improper posterior distribution. The parameter estimates and credible intervals generated in such cases have no valid probabilistic interpretation. Tak and coauthors describe this as an insidious feature of posterior impropriety, since it may go entirely undetected in practice. If nearly one-third of Bayesian analyses in leading journals have not verified the mathematical validity of their posteriors, then a meaningful fraction of the credible intervals on which astrobiological conclusions rest are formally unjustified. Frequentist procedures do not share this structural vulnerability. The validity conditions for frequentist machinery (e.g., confidence intervals, p-values, etc.) are transparent, testable, and guaranteed by construction. The Bayesian framework places the burden of validation entirely on the analyst, and the documented failure rate is not a reason for complacency.

A further structural critique concerns the concept of severity in testing: if a hypothesis passes a test severely if the data accord with it and with high probability, the test would have produced inconsistent results had the hypothesis been false. Severity is a post-data, prior-free measure of evidential weight directly assessing whether the test had genuine capacity to detect the hypothesis’s truth. Bayesian posterior probabilities provide no analogous measure. For instance, a posterior of 0.95 for the hypothesis that $f_\ell > 0.01$ may reflect a prior of 0.9 weakly updated by a single observation, or a prior of 0.1 updated dramatically by many independent measurements. The posterior probability cannot distinguish these cases, and the distinction is precisely what an evidential approach to inference requires. In a field where the temptation to import prior beliefs is greatest, because the data are few and the stakes of the questions are, arguably, enormous, severity provides the discipline posterior probabilities lack. A frequentist who reports that Earth’s biogenesis record provides insufficient statistical power to reject the hypothesis $f_\ell < 10^{-6}$ is communicating something precise, testable, and not contingent on any prior assumption. Conversely, a Bayesian who reports a posterior credible interval for f_ℓ is reporting a prior, updated negligibly, and presenting the result as a conclusion about the universe. The severity framework does not follow this misconflation because it follows directly from

the sampling distribution of the test statistic under the hypothesis.

It is worth noting the failures documented are not arguments against Bayesian methods in astronomy, astrophysics, and astrobiology in general. In settings with informative likelihoods and abundant data, Bayesian methods are powerful and appropriate, and Loredó's position is well-applied to those settings. In the regime where data are sparse relative to the dimensionality of the prior, the parameters of interest are fixed constants about which we have essentially one informative observation, and where the bounds of the inference are set by hard physical constraints rather than by sampling variation, the Bayesian framework's reliance on prior specification is, as we have argued repeatedly, a laundering mechanism through which prior belief masquerades as an empirical conclusion.

5 Conclusion

Our Fixed Observer hypothesis is the synthesis of three arguments we use to motivate the frequentist position. From the philosophy of statistics, the conditional inference tradition establishes probability attaches to procedures conditioned on the observed context rather than to fixed physical constants, the epistemic-aleatory distinction must be maintained for inference to be honest about the nature of its uncertainty, and that background scientific knowledge (e.g., light cones, cosmic horizons, conditions for biological existence, etc.) ought to structure our hypotheses rather than populate our priors. The light cone and our temporal supplement imposes a hard and permanent boundary on what any observer at our location can ever measure and renders the Drake factors fixed constants. The anthropic selection effect from Carter is better understood as a structural constraint on the valid inference class than as a probabilistic event to be incorporated via Bayes' Theorem. From the empirical record of Bayesian practice in the field, the dominant quantitative treatments of the Drake equation are systemically driven by prior specification rather than data, posterior impropriety appears in a substantial fraction of published analyses, and the severity standard cannot be met within a framework whose conclusions are prior-inherited rather than data-extracted.

These three arguments are not independent positions. The light-cone constraint is the physical realization of the epistemic-aleatory distinction: our uncertainty about the Drake factors is epistemic, arising from limited observations of fixed quantities, not aleatory, and the frequentist framework is the one which preserves this difference. The anthropic selection effect is the domain-specific instance of the reference problem: Bayesian inference treating our existence as a random draw from some prior distribution over possible observers introduces subjectivity to the prior sensitivity Spiegel and Turner identify in single-parameter estimation. The prior-dominance documented across the literature on the Drake equation, as only one example of the problem in astrobiology, is the quantitative consequence of applying a prior-dependent framework to a problem where physical constraints, not additional data, are the only available evidence.

The universe has a definite answer to whether we are alone. Our observational situation—

fixed in space by our stellar address, fixed in time by the cosmic brevity of our existence, bounded in reach by the speed of light—does not grant us the epistemic standing to gather all the data our light cone permits nor to report our uncertainty with the discipline that a procedural framework requires. In that regard, the frequentist framework is just as weak as the Bayesian one. What it can provide, however, is a scientific honesty and acknowledgment of the limitations of our knowledge.

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